AP CALCULUS AB

Unit 1 • Self Check 1.1 • Lessons 1-1 and 1-2 • Answer key

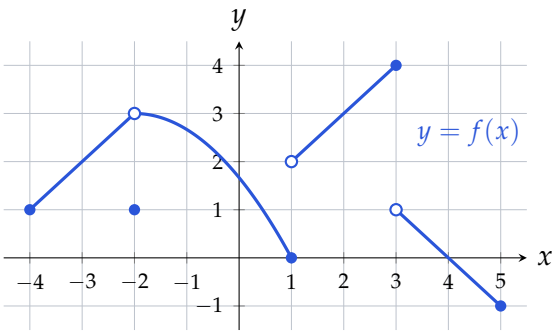
Self Check 1.1 • Lessons 1-1 and 1-2

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Questions 1 and 2 refer to the graph of f.



Question 1

NO CALCULATOR

Using the graph of f, $\lim_{x \to -3} f(x) =$

(A) 1
(B) 2
(C) 3
(D) It does not exist.

Solution

Answer: (B). On the first piece, the height at $x = -3$ is 2.

Question 2

NO CALCULATOR

Using the graph of f, $\lim_{x \to 3^-} f(x) =$

(A) 1
(B) 3
(C) It does not exist.
(D) 4

Solution

Answer: (D). From the left, the graph rises to the filled dot $(3, 4)$.

Question 3

NO CALCULATOR

$$\lim_{x\to 1}\frac{x^2+2x-3}{x-1}=$$

(A) 4
(B) 0
(C) 2
(D) It does not exist.

Solution
Answer: (A). $\frac{(x-1)(x+3)}{x-1}=x+3\to 4$.

Question 4

NO CALCULATOR

Let $j(x)=\begin{cases}2x+3, & x<-1\\ x^2, & x\geq -1.\end{cases}$ Then $\lim_{x\to -1} j(x)=$

(A) -1
(B) It does not exist.
(C) 1
(D) 3

Solution
Answer: (C). Left: $-2+3=1$. Right: $(-1)^2=1$.

Question 5 (Free response)

NO CALCULATOR

Find $\lim_{x\to 0}\frac{x^2+5x}{x}$, $\lim_{x\to -3}\frac{x^2-9}{x+3}$, and $\lim_{x\to 2} j(x)$ for $j(x)=\begin{cases}x^2, & x<2\\ 6-x, & x\geq 2.\end{cases}$

Solution
$x+5\to 5$; $x-3\to -6$; left 4, right 4: the limit is 4.